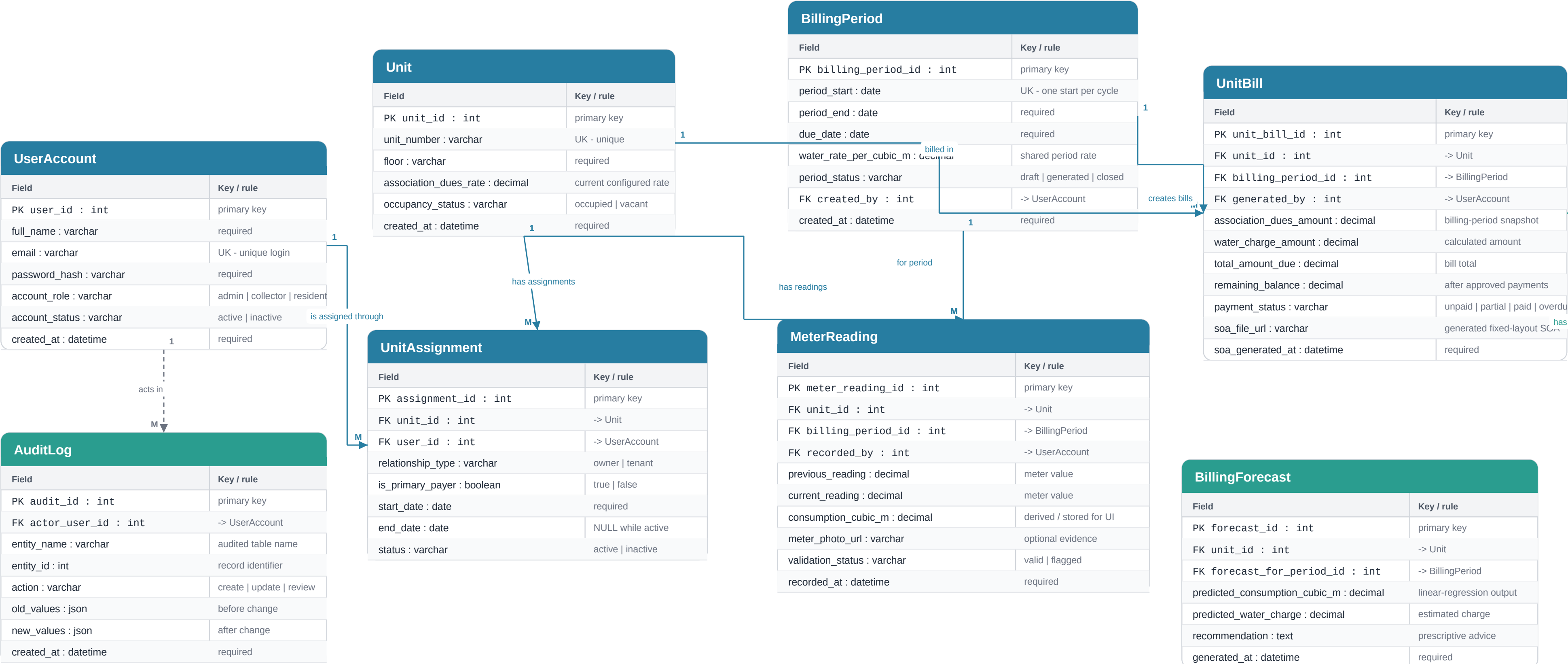


Updated ERD - Condominium Billing and Forecasting System

Simplified operational schema after removing configurable SOA templates, separate Payment records, and Notification Settings.

Legend Core table Supporting table 1-to-many relation optional/secondary relation



Important access and accountability rules

- Each owner and tenant has a separate UserAccount. They do not share login credentials.
- A resident may open a unit dashboard only when an active UnitAssignment links that user to that unit.
- UserAccount also supplies the actor FKs: BillingPeriod.created_by, MeterReading.recorded_by, UnitBill.generated_by, PaymentSubmission.submitted_by/reviewed_by, Notification.recipient_user_id, and AuditLog.actor_user_id.

Key integrity rules

- UNIQUE(UserAccount.email) and UNIQUE(Unit.unit_number).
- UNIQUE(MeterReading.unit_id, MeterReading.billing_period_id) prevents duplicate monthly readings.
- UNIQUE(UnitBill.unit_id, UnitBill.billing_period_id) prevents duplicate bills for one unit in one billing cycle.
- At most one active primary payer should exist per unit at a time.
- For Notification, at least one related source is present: unit_bill_id or forecast_id.

Scope changes incorporated

- Removed: SOATemplate. The system generates SOAs using one fixed approved layout and stores the generated file URL in UnitBill.
- Replaced PaymentProof and Payment with PaymentSubmission, which holds the receipt, OCR extraction, review decision, and verified values.
- Removed NotificationSetting. Notifications follow fixed system rules, while Notification records delivery history.
- Renamed BillingRecord to UnitBill and WaterForecast to BillingForecast for clearer business meaning.
- To keep the diagram readable, the FKs for BillingForecast and Notification are shown in their table rows rather than as extra crossing lines.